

AI Customer Support Platform

Simplified Testing Document

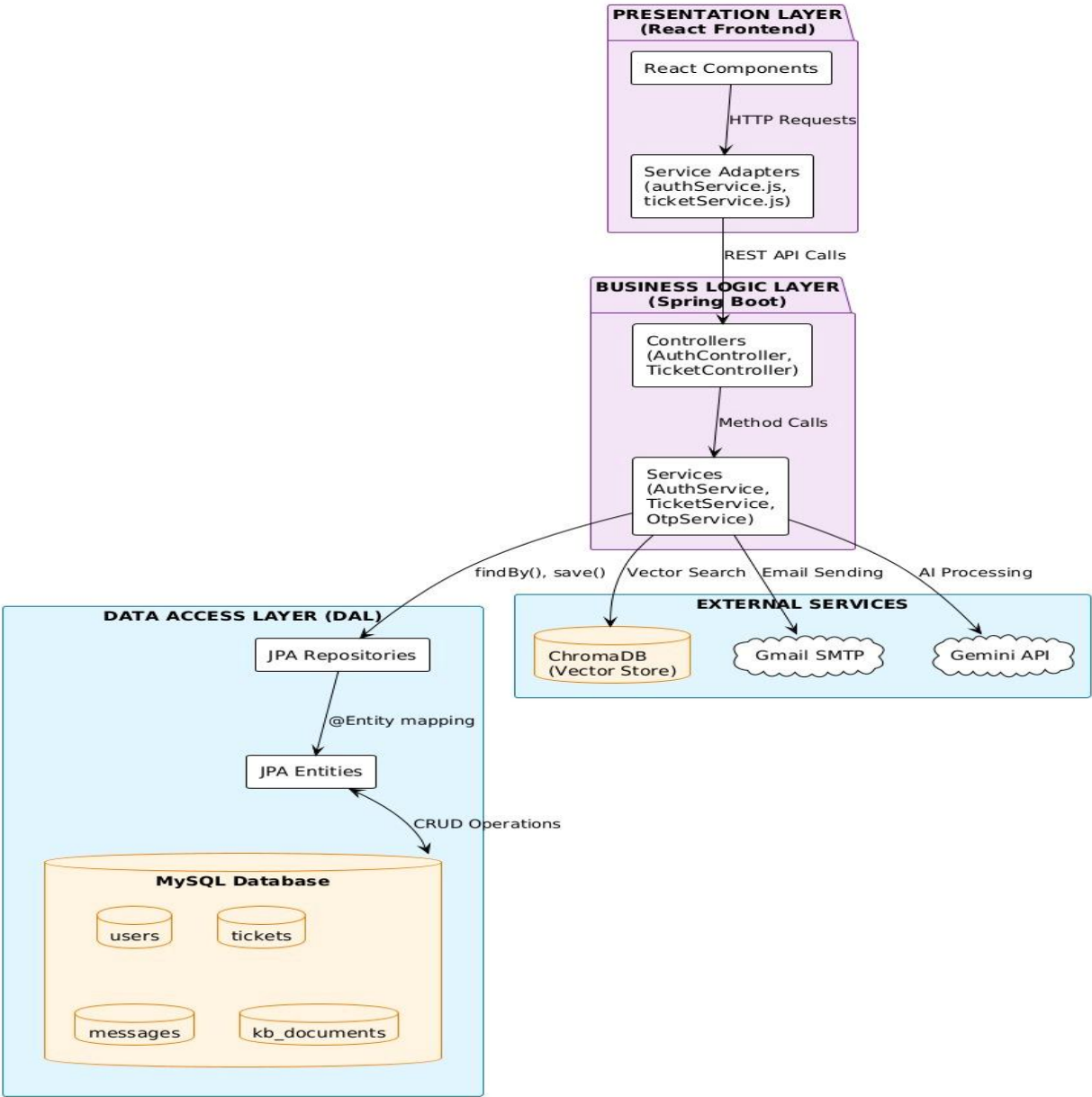
Black-Box & White-Box Testing Report | Total Tests: 91 (All Passed)

What This Document Covers

Black-Box Testing — Tests what the user sees (UI clicks, API responses, screens). No code knowledge needed.

White-Box Testing — Tests internal code logic (functions, database operations, calculations). Knows exact code paths.

Data Access Layer — AI Customer Support Platform



1. User Registration

Purpose: New users create accounts with name, email, and password.

Black-box tests UI form behavior and API responses. White-box tests internal service methods and database saves.

Test Type	What We Test	Input / Action	Expected Result
Black-Box	Form works	Valid form → Submit	Goes to OTP page
Black-Box	Weak password	Password = abc123 → Submit	Shows "Password too weak" error
White-Box	Duplicate email check	<code>register()</code> with existing email	Throws "already registered" exception

2. User Login

Purpose: Users sign in after email verification.

Black-box tests login screen and redirects. White-box tests password hashing comparison and verification flag check.

Test Type	What We Test	Input / Action	Expected Result
Black-Box	Successful login	Valid credentials → Submit	Gets JWT token, dashboard loads
Black-Box	Wrong password	Wrong password → Submit	Generic error shown
White-Box	BCrypt validation	<code>passwordEncoder.matches()</code>	Compares hash vs input

3. OTP Verification

Purpose: 6-digit code verifies email address.

Black-box tests OTP input field and success screen. White-box tests in-memory OTP store and expiry logic.

Test Type	What We Test	Input / Action	Expected Result
Black-Box	Correct OTP	Enter valid code	Email marked verified
Black-Box	Wrong OTP	Enter wrong code	"Invalid OTP" error
White-Box	Single-use logic	<code>otpStore.remove()</code> after verify	Same OTP fails second time

4. JWT Token Management

Purpose: Secure tokens for authenticated APIs.

Black-box tests API calls with/without tokens. White-box tests JWT parsing and signature validation.

Test Type	What We Test	Input / Action	Expected Result
Black-Box	Protected API	No token → API call	401 Unauthorized
White-Box	Token generation	<code>generateToken(email)</code>	Returns HS256 JWT string
White-Box	Signature check	<code>verifyWith(key)</code>	Accepts valid, rejects tampered

5. Ticket Creation

Purpose: Users submit support requests.

Black-box tests ticket form submission and list display. White-box tests ownership enforcement and database saves.

Test Type	What We Test	Input / Action	Expected Result
Black-Box	Create ticket	Logged in → Submit form	Ticket saved with ID
Black-Box	View tickets	Dashboard → My Tickets	Shows only user's tickets
White-Box	User ownership	<code>findById()</code> query	Filters by authenticated user

6. AI Chat (RAG Query)

Purpose: AI answers using knowledge base.

Black-box tests chat UI and response display. White-box tests ChromaDB similarity search and confidence calculation.

Test Type	What We Test	Input / Action	Expected Result
Black-Box	Ask question	"Reset password?" → Send	Shows answer + confidence
Black-Box	Offline AI	AI service down	Shows "AI offline" warning
White-Box	Similarity search	<code>similarity_search_with_score()</code>	Returns top 4 chunks

7. Dashboard Stats

Purpose: Shows user's ticket summary.

Black-box tests dashboard card display. White-box tests database count aggregation queries.

Test Type	What We Test	Input / Action	Expected Result
Black-Box	Load dashboard	User → Dashboard	Shows correct counts
White-Box	Count queries	<code>countByUserAndStatus()</code>	Matches database records